

Lecture Notes 8

Money and Monetary Policy

Intermediate Macroeconomics, EC2211

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Contents and literature

- Banking and financial markets
- Money
- Cryptocurrencies, stablecoins and CBDC
- The transmission mechanism of monetary policy
- Frictions in the transmission mechanism
- Inflation targeting and independent central banks
- Inflation

Literature:

- Jones (2024), chapters 10 and 14 (and some material from chapters 8-13)
- McLeay, Radia and Thomas (2014), "[Money in the modern economy: an introduction](#)"
- Royal Academy of Sciences (2022) "[The central role of banks in financial crises](#)"

Introduction

- So far, we have assumed that the central bank can set the interest rate to achieve a target level of inflation
- Today we will learn how central banks pursue monetary policy in greater detail
- Start with a primer on finance, money and the monetary system

Banking and financial markets

The financial system

1. Central banks (CB)
 - Federal Reserve (US)
 - Bank of England (UK)
 - ECB (Eurozone)
 - Riksbank (Sweden)
2. Commercial banks
 - Borrow from (or deposit money with) the central bank
 - Lend to households and firms
 - Accept deposits (create money!) from households and firms
3. Non-bank financial institutions
4. Governments, national debt offices

TABLE 10.3

A Hypothetical Bank's Balance Sheet (billions of dollars)

Assets		Liabilities	
Loans	1,000	Deposits	1,000
Investments	900	Short-term debt	400
Cash and reserves	100	Long-term debt	400
<i>Total assets</i>	2,000	<i>Total liabilities</i>	1,800
		<i>Equity (net worth)</i>	200

Table 10.3 in Jones

Commercial banks

- **Maturity transformation:**
 - Items on assets side (loans) are often committed for long term
 - Items on liabilities side (deposits) are often available on demand
- **Leverage:**
 - Equity ("net worth", "capital") is often a small fraction of the balance sheet
- Much economic activity relies on the banking system:
 - Lending to households and businesses
 - Payments
 - Wealth, savings
- But it is a fragile system
 - Tightly regulated (capital and liquidity requirements)
 - Deposit insurance
 - Bail-outs(?)

Bond prices and interest rates

Consider a bond that pays no dividends (a "zero-coupon bond")

- Suppose it is a one-year bond with nominal value $N = 100$
 - This means that the owner of the bond will receive 100 one year from now
 - Let $P_{t,1}$ denote the price of this bond at date t
 - Let $i_{t,1}$ denote the one-year interest rate at date t
 - Then

$$1 + i_{t,1} = \frac{N}{P_{t,1}}$$

- For example: if $P_{t,1} = 98$ then $1 + i_{t,1} = 1.0204$ so $i_{t,1} \approx 2\%$. You can either pay 98 to buy the bond and get 100 one year later or save 98 and get $(1 + r_{t,1})98 = 100$ one year later.
- Bond prices are thus high when interest rates are low and vice versa

Bond prices and interest rates

Consider again a bond that pays no dividends (a "zero-coupon bond")

Suppose it is a two-year bond with nominal value $N = 100$

- This means that the owner of the bond will receive 100 two years from now
- Let $P_{t,2}$ denote the price of this bond at date t
- Let $i_{t,2}$ denote the two-year interest rate (annualized) at t
- Then (ignoring uncertainty etc)

$$(1 + i_{t,2})^2 = \frac{N}{P_{t,2}}$$

Bond prices and interest rates

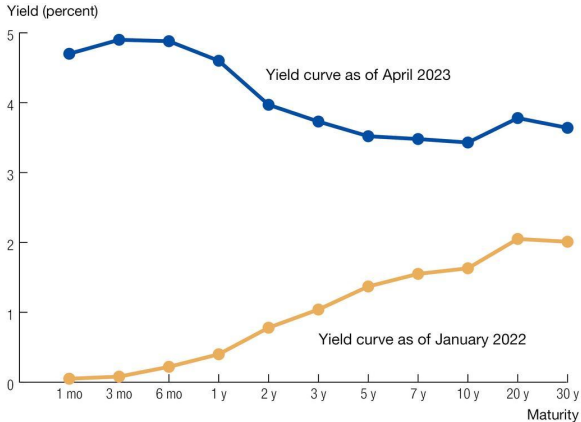
- The **yield curve** is a plot of the **term structure of interest rates**
 - It plots interest rates on same-quality bonds with different maturities (with maturities on the x-axis)
 - In our example, a plot of $i_{t,1}$, $i_{t,2}$, $i_{t,3}$, ... at a certain date t
- According to the **expectations theory** of interest rates,

$$(1 + i_{t,2})^2 = (1 + i_{t,1})(1 + i_{t,t+1,1})$$

where $i_{t,t+1,1}$ denotes the one-year interest rate at date $t + 1$ as it is priced (expected) at date t

- The **forward curve** is a graph that shows the market pricing (the expectations) today (at date t) of future short-term interest rates
 - It plots $i_{t,1}$, $i_{t,t+1,1}$, $i_{t,t+2,1}$, ... (with time at the x-axis)

The yield curve



The yield curve shows the term structure of interest rates. In 2022, U.S. Treasury bonds with larger maturities yielded higher returns. But in 2023, the yield curve had inverted. Why?

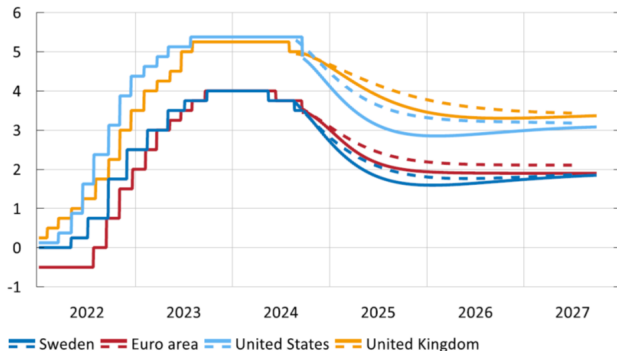
Source: U.S. Department of the Treasury, treasury.gov/resource-center/data-chart-center.

Figure 12.6 in Jones

The forward curve

Figure 8. Market expectations of policy rates in the future

Per cent



Note. The figure shows policy rates and market-based expectations according to forward pricing (Nelson-Siegel (1994)). Solid lines represent expectations on 23 September 2024. Dashed lines represent expectations immediately prior to the Monetary Policy Update in August.

Sources: National central banks and the Riksbank.

Figure 8 in Riksbank Monetary Policy Report, September 2024

Money

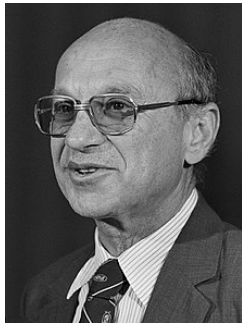
"Inflation is always and everywhere a monetary phenomenon"

-Milton Friedman

Milton Friedman

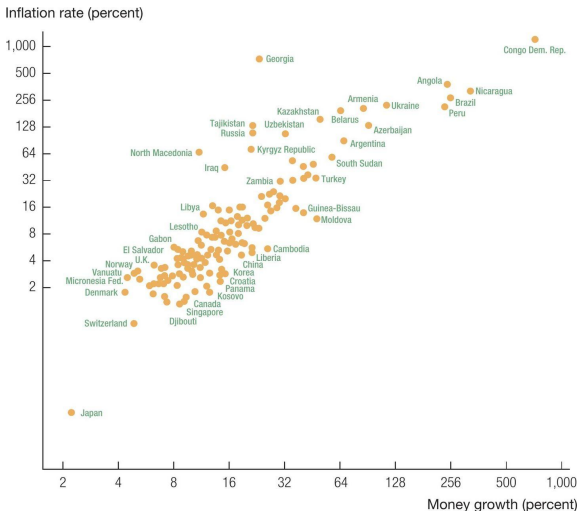
- Monetarism
- Inflation expectations
- Permanent income hypothesis
- ...

Nobel prize 1976 "for his achievements in the fields of consumption analysis, monetary history and theory and for his demonstration of the complexity of stabilization policy"



Milton Friedman
1912-2006

Money and inflation



Source: World Bank, *World Development Indicators*. The figure plots the average annual percentage change in the price level (inflation) and money supply for a sample of countries. Ratio scales are used for both axes.

What is money?

Ok, so inflation is always a monetary phenomenon. But what is money?

1. Medium of exchange
 - We use money for payments
 - Contrast: barter economies that rely on *double coincidence of wants*
2. Unit of account
 - Most prices are quoted in the same unit (the monetary units)
3. Store of value
 - A good money preserves value over time (but money is typically not the ideal storage of value)

Where does money and its value come from?

- Historically: **commodity money**
 - Goods or assets that would be valuable for other purposes are used as money (gold, silver, copper, tobacco, ...)
 - Eventually, claims on such commodities (certificates of deposits, IOU:s, banknotes) used for payments
- Today: **fiat money**
 - Money has no intrinsic value
 - A one dollar note is a claim on the Fed / government to pay one dollar
 - Money in modern economies is a *convention*, its value and usability builds on *trust and expectations*

Where does money and its value come from?

- Money is mostly created by commercial banks
 - Bank lending results in deposits
 - Deposits in transactions accounts = dominating part of "money"
 - This is not central bank money
- Central banks control the *monetary base*
 - Commercial banks avoid exposure against other banks
 - Liquidity from central banks is therefore fundamental for financial system
 - The size of the monetary base need not affect commercial bank money creation
 - But central banks provide trust through settlement in central bank money

Example economy

Consider this economy with two banks (SEB and Swedbank) and two persons (Alice and Bob)

SEB			
Assets		Liabilities	
Riksbank reserves	10	Alice's deposits	190
Bonds	30	Equity	10
Mortgage lending	160		

Swedbank			
Assets		Liabilities	
Riksbank reserves	10	Bob's deposits	190
Bonds	30	Equity	10
Mortgage lending	160		

- One morning Alice notifies SEB that she has bought Bob's house and wants to immediately transfer 20 from her account to Bob's account in Swedbank
- What can SEB now do?
- What if Alice wants to immediately transfer 50 to Bob's account?
- What if Alice just wants to *borrow* 50 and deposit in her SEB account?

Example economy

- SEB will ask the Riksbank to transfer 10 from the SEB to the Swedbank accounts in the Riksbank
- To cover the remaining 10 or 40, options are:
 - try to borrow over night from Swedbank or some other bank in the *interbank market*
 - use bonds as collateral and borrow from the Riksbank's lending facility
 - not transfer the money immediately to Swedbank – withdraw immediately from Alice's account, available in Bob's account one or a few business days later (*instant payments/settlement* is becoming more and more common, but still not broadly available for large, unexpected payments)
- If Alice wants to *borrow* 50, SEB can easily accomodate that by just changing in its own books
 - The limitations on how much "money" SEB can generate mostly come from regulations on how large its balance sheet can be relative to its equity

Money concepts

- **Monetary base (M0, MB):** Currency in circulation + reserves held by commercial banks as central bank deposits (in Sweden including Riksbank certificates)
- **M1:** Currency in circulation + overnight deposits
- **M2:** M1 + deposits with agreed maturity up to two years and deposits redeemable at notice of up to three months
- **M3:** M2 + repos + money market funds shares + debt securities with maturity up to two years

Some clarifications

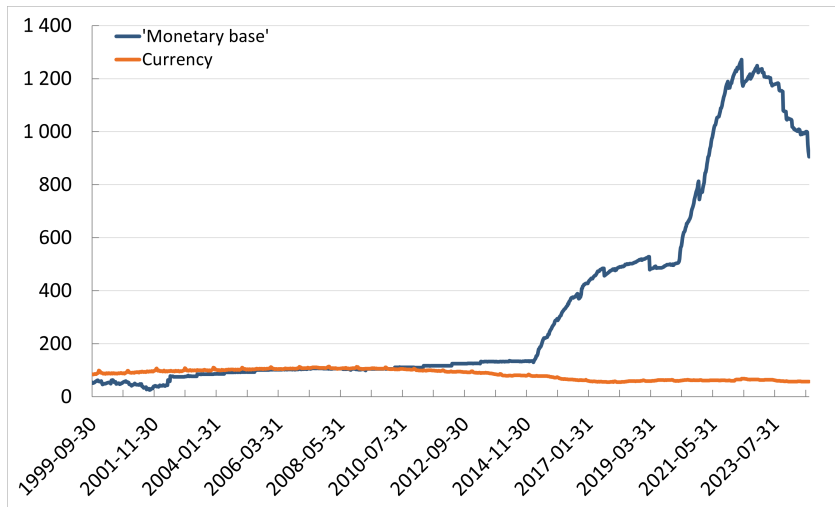
- Jones writes in chapter 8 that reserves are "accounts that private banks hold with the economy's central bank, which pay no interest". This used to be the case in the United States. Before 2008, the Federal Reserve was not allowed to pay interest on reserves. But today, paying interest on reserves is a key element of the implementation of monetary policy. And the Riksbank has been paying interest on reserves since many decades.
- Precisely how monetary policy is implemented varies between central banks. In other words, *operating frameworks* differ.
- In its policy meetings, the Riksbank's Executive Board decides on a number of different interest rates but focus the communication on the *policy rate*, previously referred to as the *repo rate*.
- The Riksbank's policy rate is the interest rate paid on Riksbank certificates. These certificates are debt instruments issued by the Riksbank weekly with a one-week maturity, and sold to its counterparties (basically the large and medium-sized banks).
- A strict definition would not include the Riksbank certificates in the measure of the monetary base (since they are not payable on demand but have a one-week maturity). In the official Swedish statistics they are nevertheless classified as part of the monetary base.
- The Riksbank also takes in deposits from its counterparties. These deposits are available to the banks on demand and pay an interest rate that is 10 basis point (0.1 percentage points) below the policy rate.
- The Riksbank also has a standing offer to lend money to the counterparties (against collateral). The lending rate is 10 basis points above the policy rate if the highest level of collateral is provided (government bonds) and 75 points above the policy rate if other high-quality collateral is provided.
- Banks regularly hold both deposits in Riksbank accounts and Riksbank certificates (see next page), but it is not common that they borrow in the lending facility.

The Swedish monetary base

	Amount	Interest rate
Currency	57	0
Deposit facility	82	3.15
Riksbank certificates	766	3.25
Lending facility	-0.02	3.35-4.00

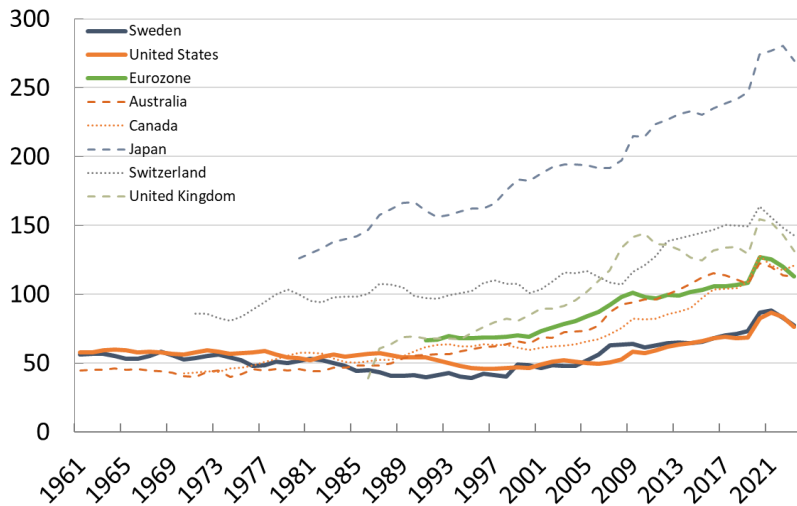
Billion SEK and percent, as of October 7, 2024. Source: The Riksbank

The Swedish monetary base



Billion SEK. Own calculations based on Riksbank weekly balance sheet. The 'monetary base' here includes certificates and deducts refinancing operations and balances on the lending facility

Broad money around the globe



M3, percent of GDP. Source: OECD

Cryptocurrencies, stablecoins and central bank digital currencies

Is bitcoin money?

Let us begin with some concepts:

- A **cryptocurrency** (or crypto asset) is a private sector digital asset that depends primarily on cryptography and distributed ledger or similar technology
- A **stablecoin** is a cryptocurrency whose value is linked to – and ideally backed by – an underlying fiat currency
 - Tether (USDT) and USD Coin (USDC) are the largest, both linked to the U.S. dollar
- An **unbacked cryptocurrency** is a cryptocurrency that is not a stablecoin
 - Bitcoin (BTC) and Ethereum (ETH) are the largest
- A retail **central bank digital currency (CBDC)** is a digital form of central bank money that can be held by households and other agents for everyday payments

Stablecoins

- To maintain a stable value against the reference asset (the U.S. dollar):
 - A stablecoin must be backed by safe and liquid reserves – this is like bank deposits without deposit insurance!
 - Reserves should be very similar to U.S. dollars
 - Stablecoin issuers have not always been transparent about what reserves they hold
 - Less regulated than banks, but now the [GENIUS act](#) in the U.S. and the [MiCA regulation](#) in the EU
- The business model of a stablecoin issuer:
 - Issue coins that do not pay interest, hold corresponding assets (e.g. U.S. Treasuries) that pay interest
- A stablecoin is similar to a bank deposit or holdings in a money market fund (but paying no interest)
- ... but may facilitate other crypto transactions

Stablecoins: macro consequences and stability risks

- Regulation and jurisdiction not always clear & no deposit insurance
 - Clear risk that there are runs on the coin
- What if there is a run?
 - Do they really hold reserves that can be liquidated quickly?
 - Would the market then be able to absorb the "fire sales" of these assets?
- Reserve accumulation implies larger demand for the reference currency
 - Easier for the U.S. government to issue new debt
 - ... "exorbitant privilege"?
 - ... or just facilitating unsustainably high U.S. debt?
- ...

While stablecoins are just a digital (tokenized?) representation of an existing currency, Bitcoin is more similar to a new currency:

- You can think of its USD price as its exchange rate against the dollar
- It is a "currency" without a central bank – its supply is instead fixed by an algorithm (at least
- It is not "money" since it is not widely used for payments, not used as a unit of account and not a particularly reliable store of value
- It would be problematic if it became "money", in particular if it was widely used as a unit of account so that firms set wages and prices in bitcoins
 - There would be a joint global monetary policy decided by an algorithm
 - There would be no lender of last resort backing up the financial system
 - ...

Central bank digital currency

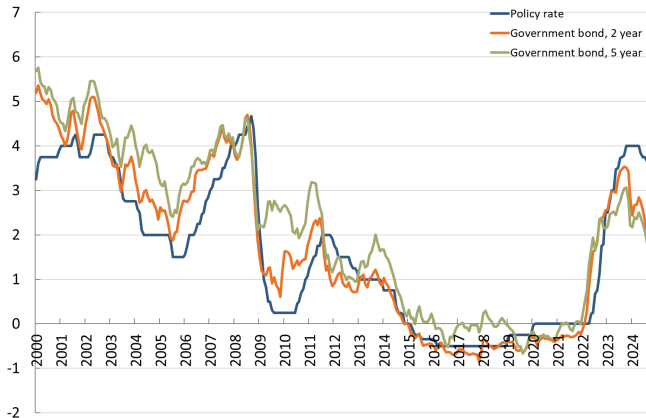
- Similar to stablecoins but issued by states / central banks
- Tokenized (retail or wholesale) CBDC could potentially facilitate digital transactions, in particular wholesale transactions between financial agents
- In the euro area it could be a substitute for non-existing pan-European digital payments systems
- A potential alternative is **tokenized bank deposits**

The transmission mechanism of monetary policy

Monetary policy in normal times

- The central bank decides on a (very!) short interest rate
- This interest rate may in itself be relatively unimportant
- The transmission mechanism of monetary policy:
 - Interest-rate channel: Expectations theory of interest rates → effect on longer interest rates → investment and consumption
 - Cash-flow channel
 - Banking channel
 - Expectations channel, communication

Transmission to government bond rates



Percent. The Riksbank's policy rate and interest rate on government bonds, monthly averages. Source: Riksbank and Statistics Sweden.

Transmission to interest rates faced by households and firms

Per cent

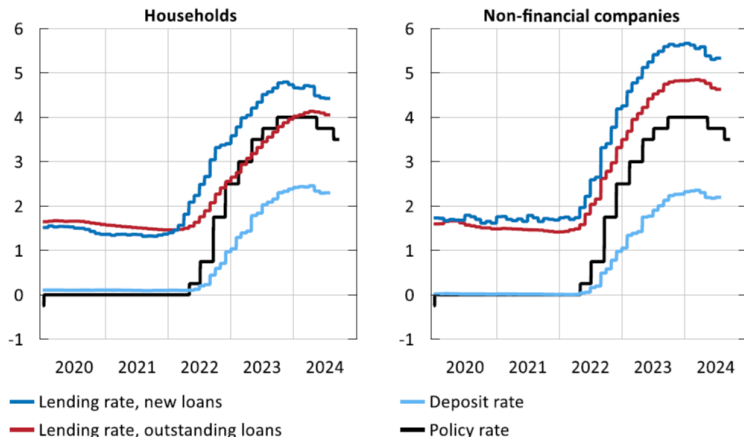
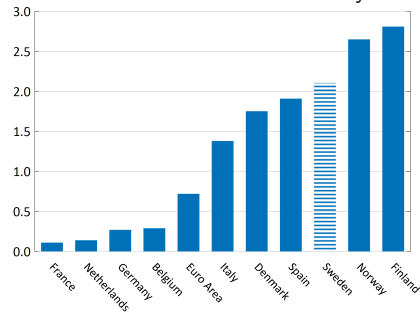


Figure 10 in the Riksbank's [Monetary Policy Report](#), September 2024

Cash-flow effects

- If interest rates are variable and mortgages large ...
- ... monetary policy has large effect on cash-flows

Increase in interest rate paid by households from Dec. 2021 to July 2023



Percentage points. Source: Macrobond.

Banking channel

- Balance-sheet channel: When interest rates fall, asset prices increase and firms have more investment opportunities
 - Firms have more collateral
 - Banks judge that firms have more profitable investment opportunities
- Bank-lending channel (or bank-credit channel): When interest rates fall, bank assets (long-term lending) increase in value
 - Banks then tend to have more equity
 - More equity means more lending capacity

Expectations channel

- Arguably the core of modern central banking
- Recall the importance of:
 - Inflation expectations in the Phillips curve
 - Interest-rate expectations in the transmission of policy rate to longer-term rates
 - The need for a nominal anchor in the economy: something to provide the basis for wage- and price-setters
- A credible inflation target (or some other nominal anchor) → more efficient and stable economy, more flexibility for monetary policy
- How to build credibility? How to communicate effectively?

(More on this in section on independent central banks)

The effects of monetary policy

How do we use data to analyze the effects of monetary policy?

Why would it be problematic to just look at the correlation between the central bank's policy rate and inflation?

The central bank raises the policy rate when inflation is high!

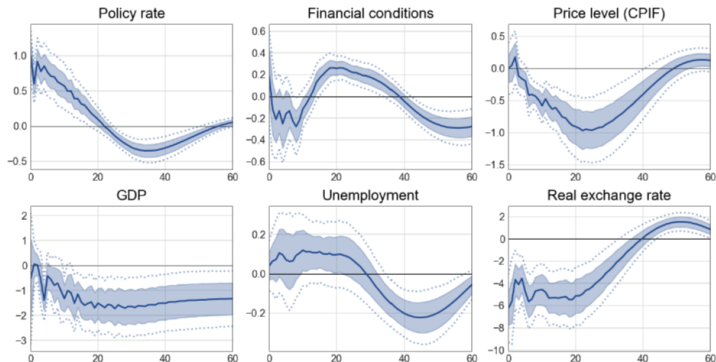
- The policy rate may be positively correlated with inflation
- ... although we think that high policy rates lead to low inflation

To identify causality from policy rate to economic outcomes such as inflation and output, we must use clever identification strategies and/or suitable econometric techniques

- Common identification method: consider effects of monetary "surprises"
- These are arguably unrelated to the current state of the economy

The effects of monetary policy

Figure 4: Effect on macroeconomic variables of a target factor shock



Note: Data is monthly. The policy rate is measured at its end-of-period value. The solid lines show the point estimates of the impulse response functions following a target factor shock. The impulse responses are normalised to yield a 1 pp. higher policy rate on impact. The shaded areas show 68 per cent bootstrapped confidence intervals around the point estimates and the dotted lines show the edges of the 90 per cent confidence intervals. The policy rate and unemployment rate are measured in percentage point deviations. Financial conditions are measured in index point deviations. The rest of the series are measured in per cent deviations.

**Frictions in the transmission
mechanism:**

**Monetary policy when market
conditions are less favorable**

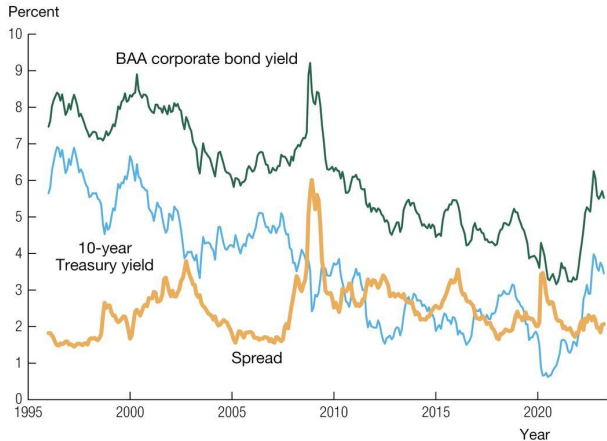
What could possibly go wrong here?

An incomplete list:

- Poorly capitalized and/or not profitable banks
 - May have to deleverage, stop lending
 - Or just go bankrupt
 - Subject to *bank runs*
- Other financial-market disruptions
 - Increasing risk premia
 - Fire sales
- Nominal interest rate hitting the lower bound
 - Interest rate cannot be (much) below zero
 - ZLB: zero lower bound; ELB: efficient lower bound
- Loss of central bank credibility
 - Rising inflation expectations
 - Volatile expectations of central bank's policy rule

Increasing risk premia

Despite the large decline in the interest rate on government securities, the interest rate for corporate borrowing rose rather than declined during the financial crisis.

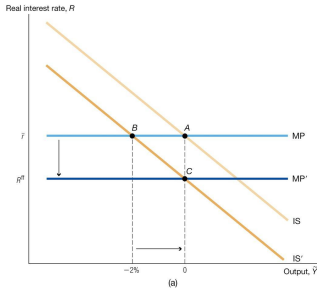


Source: The FRED database, series BAA, GS10, and author's calculations.

Figure 14.1 in Jones

Financial crisis with increasing risk premia in the GFC

(a) The collapse of a housing bubble causes the IS curve to shift back, producing a recession at point B. In response, the Fed lowers the fed funds rate, pushing the economy back to its potential at point C.



(b) The financial crisis raises interest rates despite the Fed's efforts, producing a deep recession at point D.

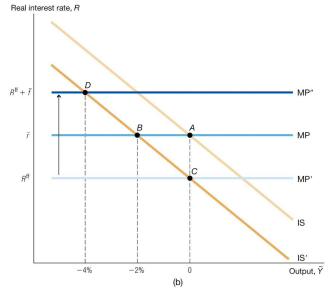
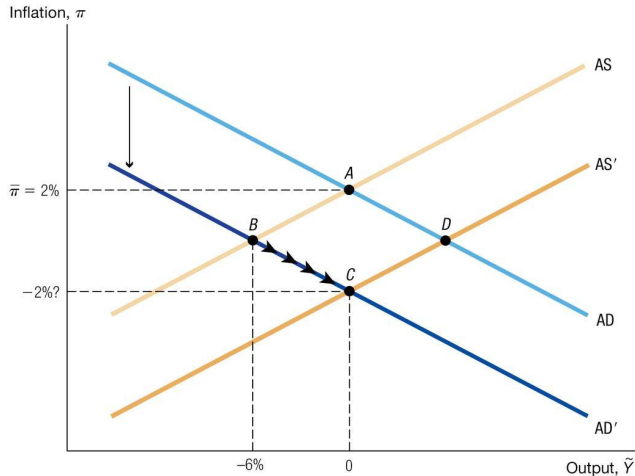


Figure 14.2 in Jones

Financial crisis with increasing risk premia



The large negative AD shocks associated with the financial crisis cause a deep recession, and the economy moves from A to B. The recession causes inflation to fall over time, shifting the AS curve down and moving the economy gradually to point C. Because inflation was already relatively low at 2 percent before the financial crisis, this decline could potentially lead to deflation.

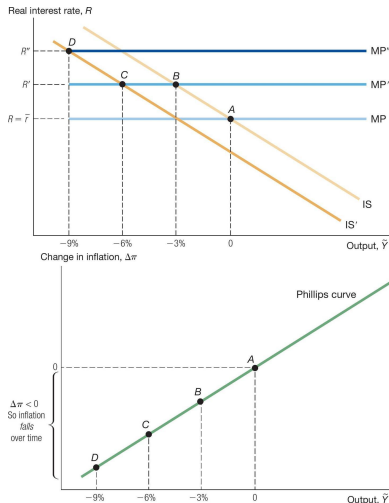
Figure 14.3 in Jones

1930s Great Depression

- Wide-spread bank failures (1000s of banks went bankrupt)
- Milton Friedman and Anna Schwarz (1963), *A Monetary History of the United States*:
 - The Fed was too passive/slow/incompetent after 1929 stock market crash
 - Too restrictive monetary policy, let money supply fall
 - As a result, deflation (negative inflation) and long, deep recession
- Ben Bernanke (1983):
 - The core problem was that bank runs and bank failures resulted in a severe *credit crunch* that exacerbated the crisis
- Moreover, *gold standard* put restrictions on monetary policy

1930s Great Depression

- See Jones, exercise 14.2
- Tightening of U.S. monetary policy in 1928-29 (from A to B)
- Stock-market crash, loss of wealth (from B to C)
- Prices are falling, but Fed keeps nominal interest rate unchanged ... so the real interest rate is increasing (from C to D)



What can central banks do?

- Lender of last resort
 - Walter Baghehot (1873): Lend freely to solvent depository institutions against good collateral

But ...

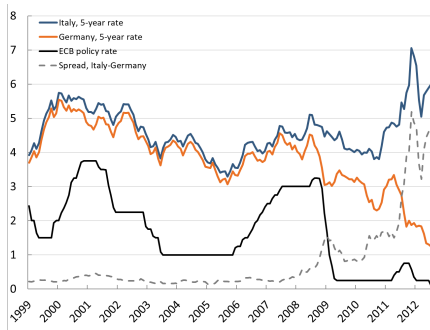
- In a crisis, difficult to assess who is solvent and the value of collateral
 - Solvency and asset values depend on what central banks do
 - And what if the important actors are not solvent or do not have useful collateral?
- Market maker of last resort
 - Step in as buyer on important markets to stop "fire sales", to push down risk premia, to reach non-banks
- "Talk"
 - Confidence
 - Forward guidance

What can central banks do?

- In recent crises, central banks (and governments) have been much more active than in the 1930s
- Some episodes:
 - 2008 Global financial crisis
 - 2010-2012 European debt crisis
 - 2019 Repo-market turmoil
 - 2020 Pandemic dash-for-cash
 - 2022 U.K. gilt crisis
 - 2023 Banking crisis
- Some measures:
 - Asset purchases ("quantitative easing", QE)
 - Generous lending facilities (e.g. pricing, collateral accepted)
 - "Funding for lending"
 - Bailouts

The Euro crisis

- Increasing risk premia on government bonds in some countries after GFC
- "Doom-loops"?
 - If you think that country X will default, the interest rate on its bonds increase
 - With higher interest rates, it is more difficult to repay or refund debt that matures
 - Self-fulfilling expectations, good and bad equilibria

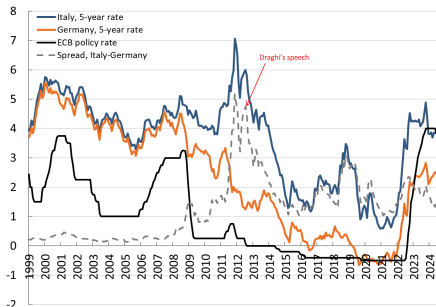


Percent (interest rates) and percentage points (spread). Source: FRED

The Euro crisis and the power of words

- Mario Draghi, then President of the ECB
- Speech in London, July 26, 2012: "We will do whatever it takes to preserve the euro. ... Believe me, it will be enough."

For a little bit more on this, see e.g. this [KfW note \(2017\)](#)

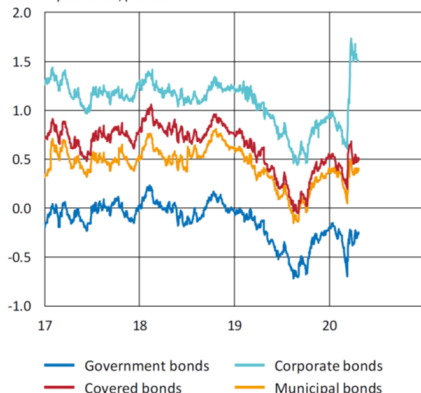


Percent (interest rates) and percentage points (spread). Source: FRED

Risk premia during covid

Figure 1:11. Bond yields

5-year bonds, per cent

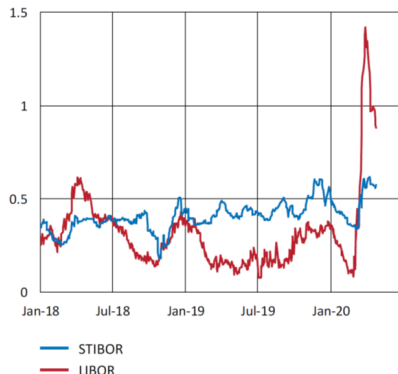


Note. Covered bonds and corporate bonds are zero coupon rates calculated using the Nelson-Siegel method. Corporate bonds for companies with credit ratings of BBB or higher. Municipal bonds are benchmark bonds, issued by Kommuninvest i Sverige AB.

Sources: Macrobond and the Riksbank

Figure 2:7. Difference between interbank rates and rates on treasury bills, 3 months

Per cent



Note. For LIBOR, the difference is taken in relation to a US benchmark treasury bill, and for STIBOR in relation to a Swedish treasury bill with a zero coupon rate calculated using the Nelson-Siegel method.

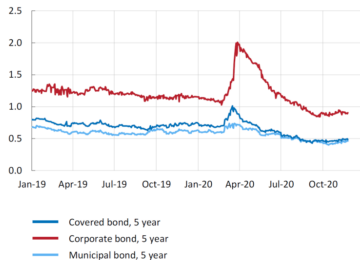
Sources: Macrobond and the Riksbank

Risk premia during covid

- March and April 2020: massive support programs
... from central banks and governments
- Ambition to be quicker and more decisive than in 2008 GFC
... and premia quickly fell back
- Pandemic recession was sharp, but much shorter than typical recessions
... a financial crisis was avoided

Figure 5. Yield differentials between different types of bonds and government bonds

Percentage points



Note. Zero coupon rates calculated using the Nelson-Siegel method. The corporate bond series illustrates a heterogeneous group of bonds for companies with credit ratings of BBB or higher. Municipal bonds are issued by Kommuninvest i Sverige AB.

Sources: Macrobond, Refinitiv and the Riksbank.

Figure 5 in Riksbank [Monetary Policy Report, November 2020](#)

March 2023 banking crisis

Silicon Valley Bank (SVB), First Republic, Signature, Credit Suisse (CS)

Losses due to higher interest rates:

- Bond values fell substantially
- Slowdown of venture capital business, crypto initiatives, ...
- Longer run structural problems in CS

Sudden runs on these banks:

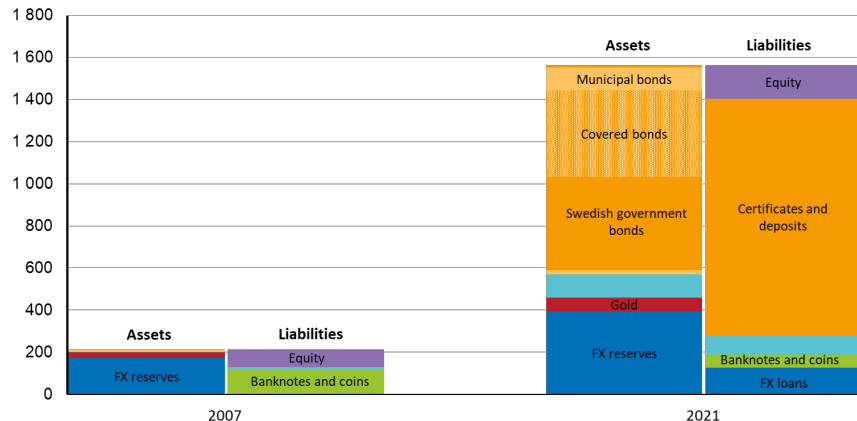
- Attempts to withdraw more than 50% of SVB deposits

"Silicon Valley Bank"			
Assets			Liabilities
Reserves	20	Deposits	170
U.S. government bonds	190	Market funding	25
		Equity	15

Policy response?

- U.S. and Swiss governments stepped in
- Examples:
 - 94 percent of SVB deposits were *not* covered by the deposit insurance
 - ... but all SVB deposits were anyway guaranteed by the government
 - Fed (temporarily) accepted all government bonds as collateral at face value even though market value was much lower

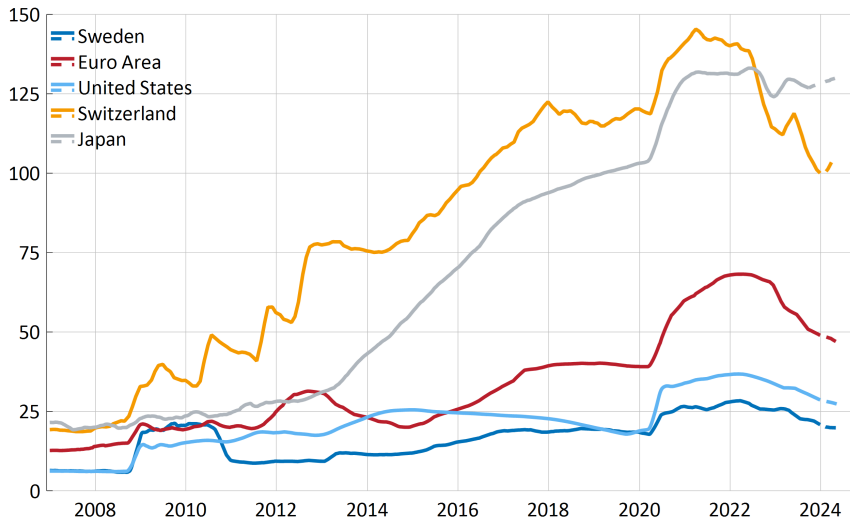
The Riksbank's balance sheet then and "now"



SEK billion

Figure 11 in Flodén (2022)

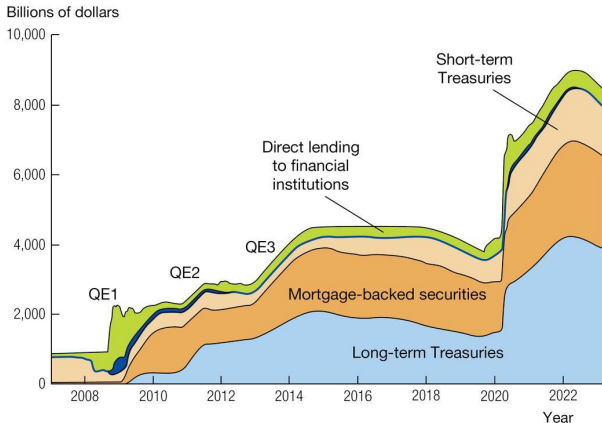
Central banks' balance sheets then and now



Central bank balance sheets, percent of GDP, own calculations.

The Fed's balance sheet then and now

Between the global financial crisis and the Covid-19 pandemic, the Federal Reserve expanded its balance sheet by a factor of 10 via unconventional monetary policy.



Source: Credit Easing. Federal Reserve Bank of Cleveland.
clevelandfed.org/en/our-research/indicators-and-data/credit-easing.aspx. Reprinted with permission.

Figure 14.9 in Jones

What are the costs of all these interventions?

- Moral hazard, excessive risk taking: "Heads I win, tails you (the tax payer) lose"
- Loss of price signals from the market
- Central banking becomes more multi-dimensional
 - More "political"
 - More difficult to square with central bank independence
- Important to strengthen the financial system in good times
 - Bank regulation
 - Macroprudential policies (e.g. relating to household debt, mortgages)
- In a crisis, intervention typically is the least of evils

Inflation targeting and central bank independence

The time inconsistency problem of monetary policy

Recall the time inconsistency problem of monetary policy from LN07

- Kydland and Prescott (1977): policy makers' inability to commit to low inflation may give rise to excessive inflation
- In the absence of a binding commitment to low inflation, inflation may increase without any corresponding increase in output

Solving the time-inconsistency problem

Set up institutions to eliminate root of the problem:

- Delegate monetary policy to independent central bank
- Formulate (politicians can do this) clear objectives, the target
- Committee with overlapping terms (avoid final-period problem)
- Communicate transparently: the target, tools used, analysis/forecasts/motivation
- All this allows the central bank to build credibility, accumulate a reputation, to not exploit the Phillips curve

Solving the time-inconsistency problem

The EU legislation (the *Treaty*), establishes independence of central banks:

- EU Article 130: "neither the ECB, nor a national central bank, nor any member of their decision-making bodies shall seek or take instructions from Union institutions, bodies, offices or agencies, from any government of a Member State or from any other body. . . . governments of the Member States undertake to respect this principle . . ."
- ECB Article 14: "... the term of office of a Governor of a national central bank shall be no less than five years. A Governor may be relieved from office only if he no longer fulfils the conditions required for the performance of his duties or if he has been guilty of serious misconduct."

Monetary policy committees

- Fed: FOMC (Fed Open Markets Committee)
 - FOMC = Board of Governors (6 persons) and presidents of the 11 regional Fed banks
 - Board of Governors: appointed by U.S. president (Senate approval), 14-year terms
 - Voting: BoG + president of NY Fed + (rotating) 4 other regional presidents
- ECB: Governing Council (GC)
 - GC = Executive Board (6 persons) and governors of the 20 national central banks
 - Executive Board: appointed by European Council (EU parliament approval), non-renewable 8-year terms
- Riksbank: Executive Board (EB)
 - EB appointed by the Riksbank's General Council for 5- or 6-year terms, renewable once

Inflation targeting

- Inflation targeting provides a *nominal anchor*
 - Coordinates expectations
 - Facilitates wage and price setting
- Mandates are for *flexible inflating*
 - Central banks shall also promote economic stability, high growth and low unemployment
 - The U.S. Fed has an explicit *dual mandate*
- Almost all developed economies have their target at **2 percent**

Inflation expectations

- Key information in monetary policy process
- How are they measured?
 - Surveys: households, firms, financial-market participants, wage setters, purchasing managers
 - Financial markets: inflation-protected bonds ("TIPS"), explicit inflation contracts ("inflation swaps")
- Stability of long-run inflation expectations was important in recent inflation episode
 - Facilitated rapid return to inflation target

Inflation expectations in Sweden

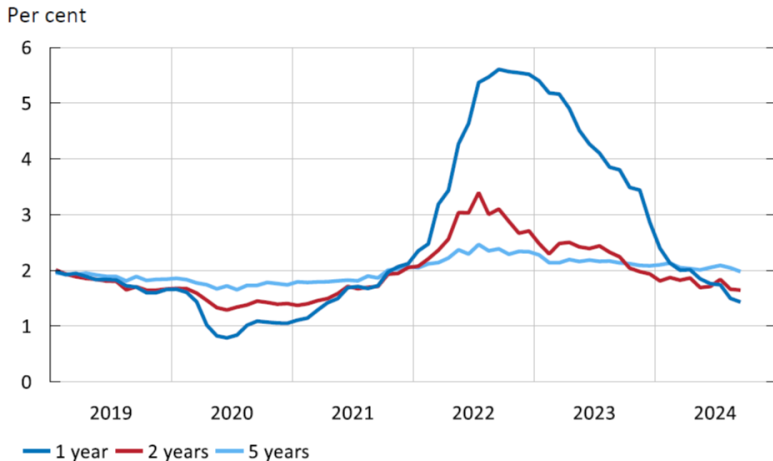


Figure 20 in Riksbank [Monetary Policy Report, September 2024](#). Refers to money market agents' expectations on CPI inflation in Kantar Prospera survey.

Inflation

The percentage change in the aggregate price level over a given time interval:

$$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}}. \quad (1)$$

Normally expressed as percentage change over 12 months

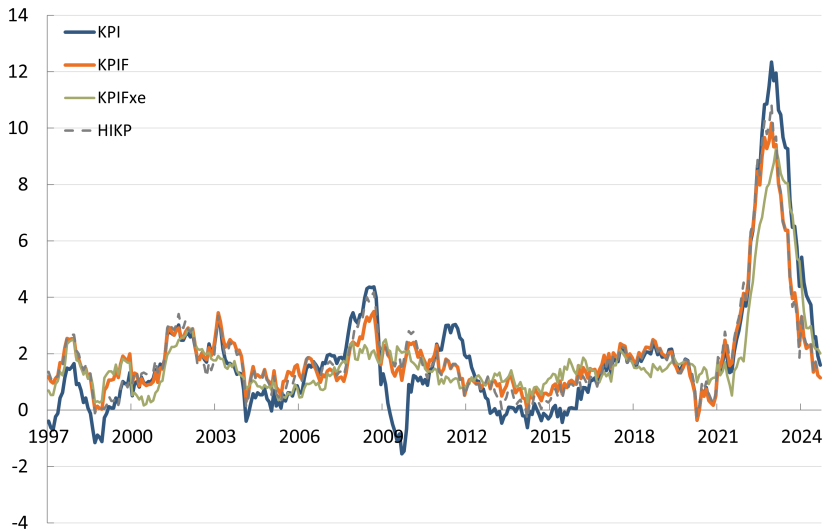
Measuring the price level

- Consumer Price Index (CPI):
 - A cost-of-living index
 - Tracks the price of a representative basket of goods over time
 - Basket updated in January every year
 - Includes energy prices, value-added taxes, rents, owner-occupied housing costs
- The CPIF
 - "CPI with Fixed mortgage rates"
 - Eliminates mortgage-rate effect from owner-occupied housing costs
 - The Riksbank's target variable: objective is to have 2 percent CPIF inflation

Measuring the price level contd.

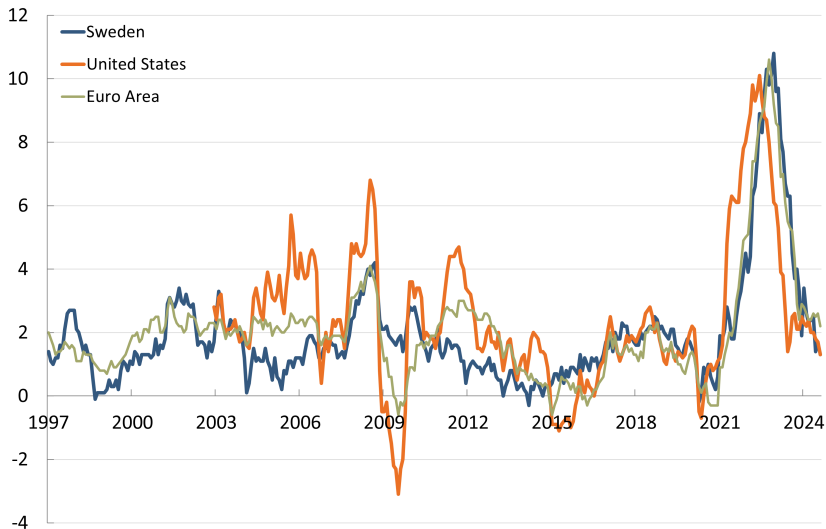
- CPIF excluding energy prices
 - Eliminates volatile energy prices
 - A measure of "core" inflation
 - Idea is to get better indication of underlying inflationary pressures, where is inflation heading?
- Harmonised Index for Consumer Prices, HICP
 - Standardized European measure (but all important details have not been standardized)
 - Currently does not include costs of owner-occupied housing
 - ECB's target variable
 - Swedish HICP is typically very similar to CPIF
- GDP-deflator
 - Measures the price level of domestically produced goods and services in the economy

Swedish inflation



Percent annual change in price index. Source: Statistics Sweden.

HICP inflation in Sweden, United States and Eurozone



Percent annual change in HICP. Source: Eurostat.

Why is high inflation a problem?

- Redistributes wealth arbitrarily across groups (young/old, borrowers/lenders)
- Makes the future unpredictable and affects incentives to invest
- High inflation may quickly spiral out of control if it induces wage increases (a wage-price spiral)
- Money becomes less effective as a *unit of account*

Why is deflation a problem?

- Problems with deflation are similar to problems with inflation: uncertainty, calculation costs, arbitrary redistribution, etc
- Recall Fisher equation ($r = i - \pi^e$) and Zero Lower Bound for the nominal rate
 - If inflation is very low (very negative!), the real interest rate will be high
 - Deflationary expectations make saving more attractive, monetary policy contractionary
 - Another way to say this: If prices are expected to fall, households may defer consumption
- Increases the value of debt in real terms
- Difficult to adjust real wages between sectors, firms or worker if there is downward nominal wage rigidity